ALGONQUIN COLLEGE

CST8507: NATURAL LANGUAGE PROCESSING

WEEK#1

COURSE OVERVIEW & INTRODUCTION TO NLP

BY
HALA OWN, PH.D.

Lesson Agenda

- Course CSI
- What is Natural Language Processing (NLP)?
- Applications of Natural Language Processing
- Challenges in Natural Language Processing
- Approaches to NLP
- NLP Libraries in Python

Contacting your Professor

- Office hours: TBA

Room#: WT315

Email: ownh@algonquincollege.com

- Expect email reply is within Two business days. Therefore, you need plan your emails accordingly
- Include in your email:
  - Your name
  - Course name and code
  - Section No.

Evaluation Breakdown

Assessment	**Value**	**CLRs**
Lab (4x 5%)	20%	1, 2,3, 4,5
Assignment (Week 6)	10%	1, 2,3, 4
Assignment 2 (Group Project)	20%	1, 2,3, 4,5,6
Total Practical	**50%**	
Midterm (1 x 15%, Week 7)	15%	1, 2,3, 4
Final Exam (1 x 25%, Week 15)	25%	1, 2,3, 4,5,6
Participation (5 out of 8)	10%	1, 2,3, 4,5
Total Theory	**50%**	

Labs (5) 15%

Lab1
Lab2
Lab 3
Lab 4

Grades are equally distributed

- Labs must be **demonstrated** according to their scheduled weeks (stated in the CSI).
- Lab professor may ask you to do something else to verify that you have done the lab by yourself.

Assignments

- **Assignment 1** (10%) **week 6: Individual work**

- **Assignment 2(Project) 20%**
    - proposal Week #9
    - Submission(Code+ Report+ Presentation) Week #12
    - group working (max. **two members**)

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Midterm Exam

- Week 7 Lecture
- 60 min
- Materials :Week 1 till Week 6
- Weight: 15%
- Closed Book
- On paper

Bring your own cheat sheet

7

Final Exam(Theory)

- Week 15
- Materials:Week1 till Week 12
- Weight: 25%
- Closed Book
- On paper

Bring your own cheat sheet

Participation (10%)…

- Five best Participation will be counted into final score. Each will occupy 2% of final score
- The content in the Participation will be based on the **hybrid** which have been posted that week and the **topic which has been covered that week**
    - On Brightspace
    - Multiple Choice / True & False
    - Open book
    - one attempt
    - Time : 10 min.
- Participation will be proctored **during the lecture**.
- Participation must be completed in person during the scheduled lecture time. Submitting a participation **without attending** the lecture will result in a grade of **zero**.

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Lecture Rules

- Feel free to **ask questions** at anytime at all. Don’t be shy, you’re not the only one who will have questions.
- Most topics covered in the **lecture slides will be demonstrated**. Feel free to follow along with the demonstrations with your own computers.
- **Note-Taking**

Lab Rules

- Attendance is mandatory for the weeks that have lab activities assigned or require you to demo your work.
- Working with others is encouraged. Explaining concepts and your way of thinking is allowed. **Copying someone's answers is not**.
- Come and go as you please, just ensure you have time to demo the lab to the instructor before the class ends.

Important Information

Passing the course

- You must get at least **50% in practical** (labs, assignments) and at least **50% in theory** (quizzes, midterm and final exam) to pass the course. Simply getting 50% overall without meeting the 50% min in each theory and practical is not sufficient to pass the course.

Important Information…

- Labs, and Assignments must be **demonstrated immediately after the due date and according to their scheduled weeks (as stated in the CSI).** Missed Demo on the specified date may result in zero mark.
- Demos will be conducted during your lab session, during which your lab professor may ask you to perform additional tasks to ensure the authenticity of your work. Failure to answer questions about your work may result in a Zero mark.

Important Information…

- The demo is not for evaluation
- Demonstrating something **that does not match what you submitted** on Brightspace would result in a zero mark.
- The official submission deadline for all labs and assignments is **Friday night (11:59 PM).** To accommodate all learners, I allow submissions until **Sunday night (11:59 PM) without any penalties**.
- **No Extension or late submission for the Assignment 2 (Project).**

Hybrid Work

- Each Week , Hybrid materials will be released
- **Part from your course**
- Additional materials related to each week’s topic

Textbooks

- Jurafsky, D. & Martin, J. H. (2024) *Speech and Language Processing (3rd ed. draft)*. https://web.stanford.edu/~jurafsky/slp3/ed3book.pdf

- Bird, S., Klein, E., & Loper, E. *Natural Language Processing with Python.* (O'Reilly 2009, website 2018), http://www.nltk.org/book/

Tools

- Jupyter Notebook
- Google Colab

Pre-Requisites

- Fundamental Knowledge of Machine Learning(CST8502)
- Fundamental Knowledge of Python

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Knowledge Representation

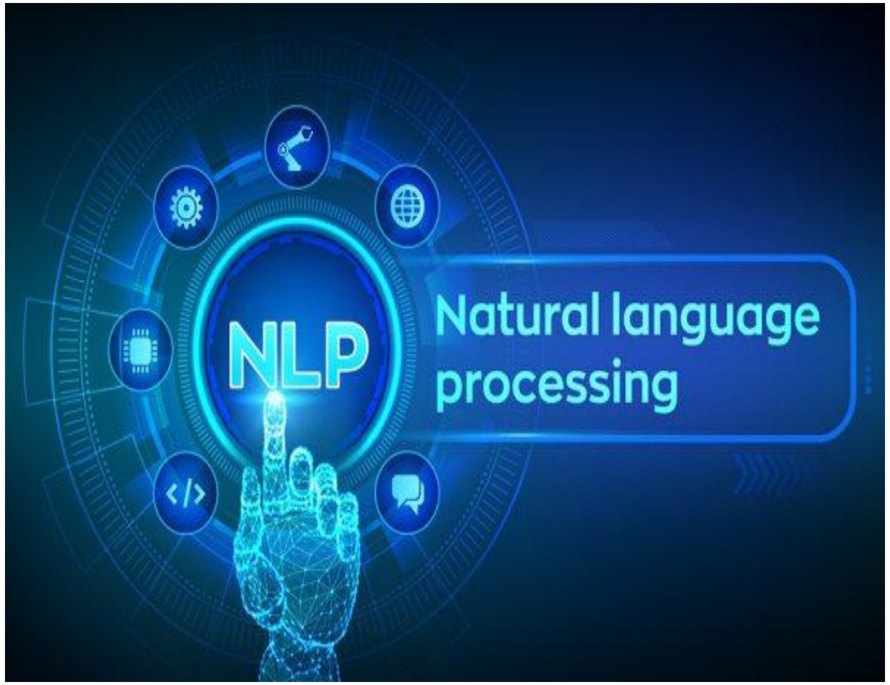
Unstructured
Ambiguous
Lots and lots of it!

Structured
Precise, Actionable
Specific to the task

Humans can read them, but
… very slowly
… can’t remember all
… can’t answer questions

Computers can use
… quickly answer questions
… memory is not a problem
… don’t get tired

What Is Natural Language Processing?

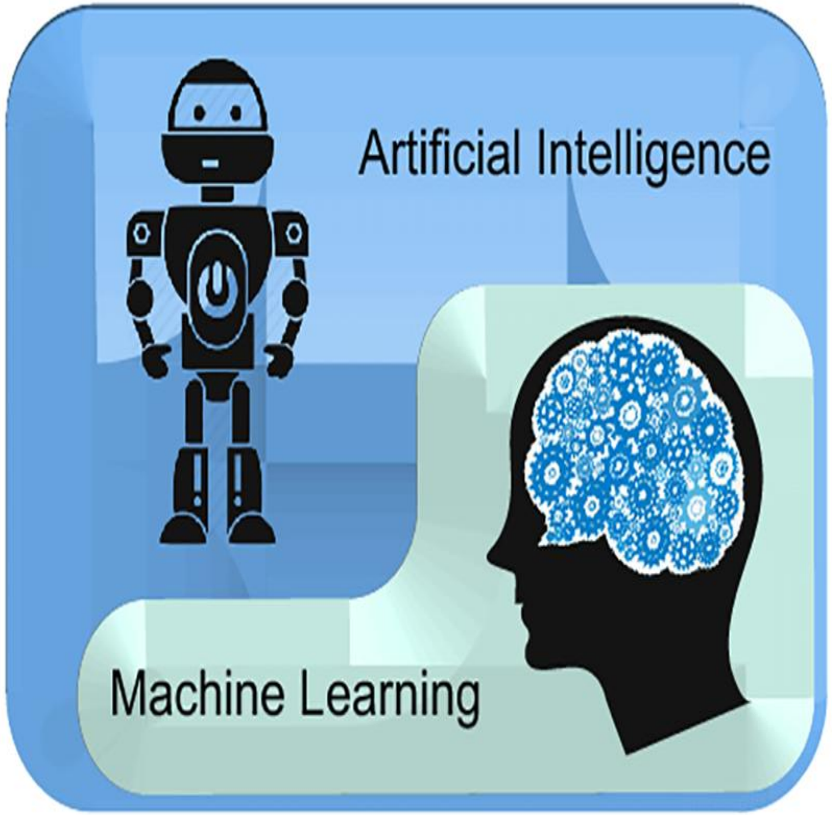


- Subfield of linguistics, computer science, and artificial intelligence concerned with the **interactions between computers and human language.**
- How to program computers to process and analyze large amounts of **natural language data.**

Motivations

- Enables computers to understand and process human language
- Helping businesses to analyze customer feedback and improve customer service
- Making it easier for individuals with **disabilities** to interact with computers and access information
- Extract insights and make sense from data

Discussion Activity



Join at
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#1046 936

Interconnections Between AI, ML, DL, and NLP

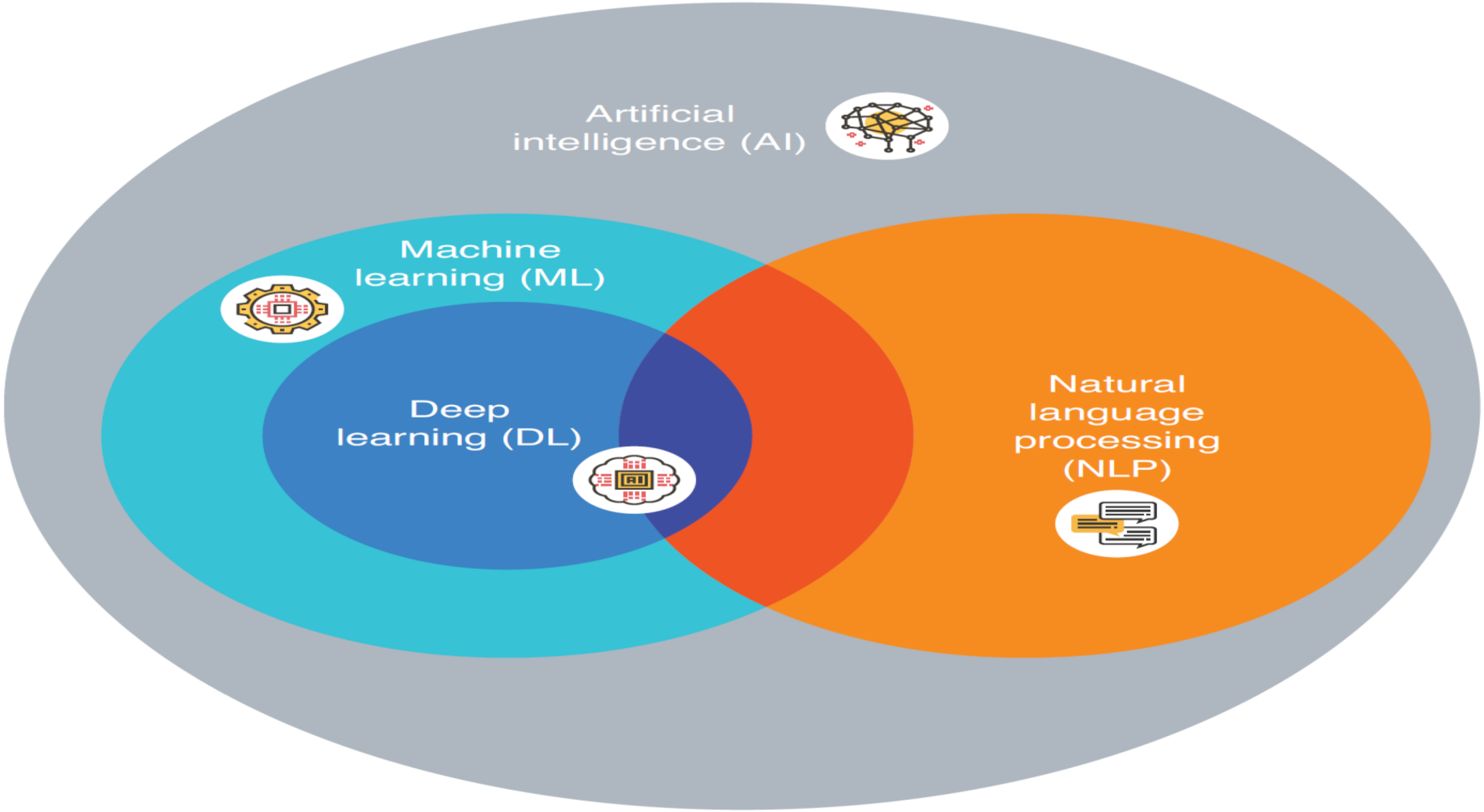


Image source: Real -World NLP, 2021 by Manning Publications Co.

History of NLP

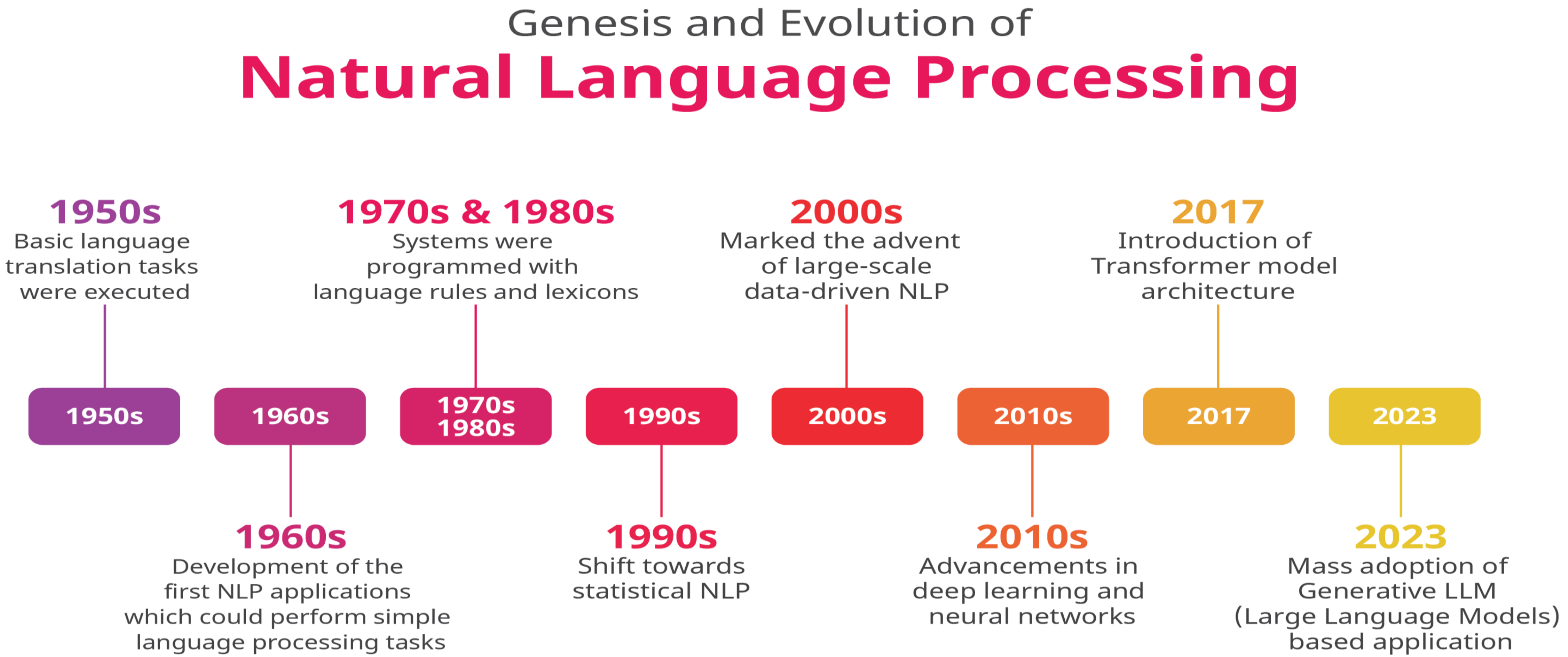
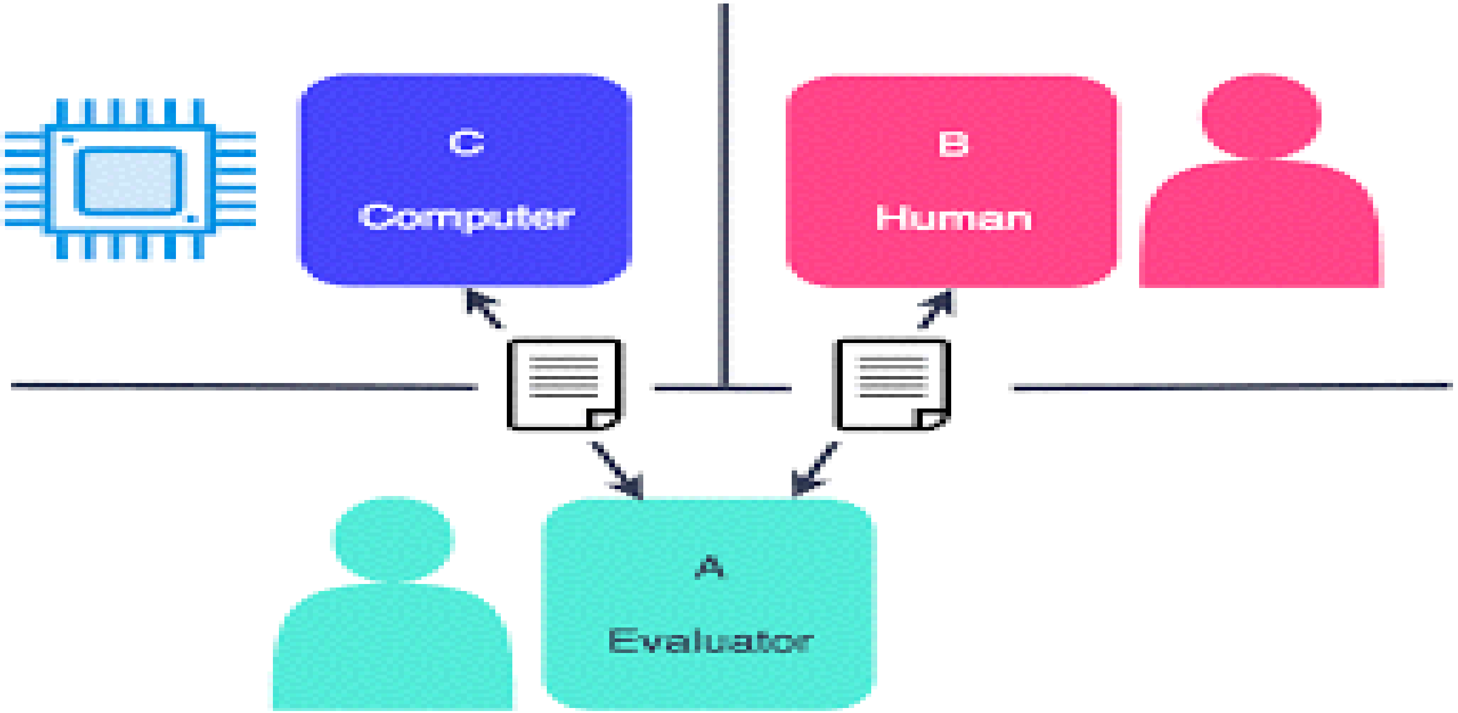


Image source: https://commtelnetworks.com/exploring-the-impact-of-natural-languageprocessing-on-cni-operations

Turing's test for Artificial Intelligence



Ability to understand and generate language ~ intelligence

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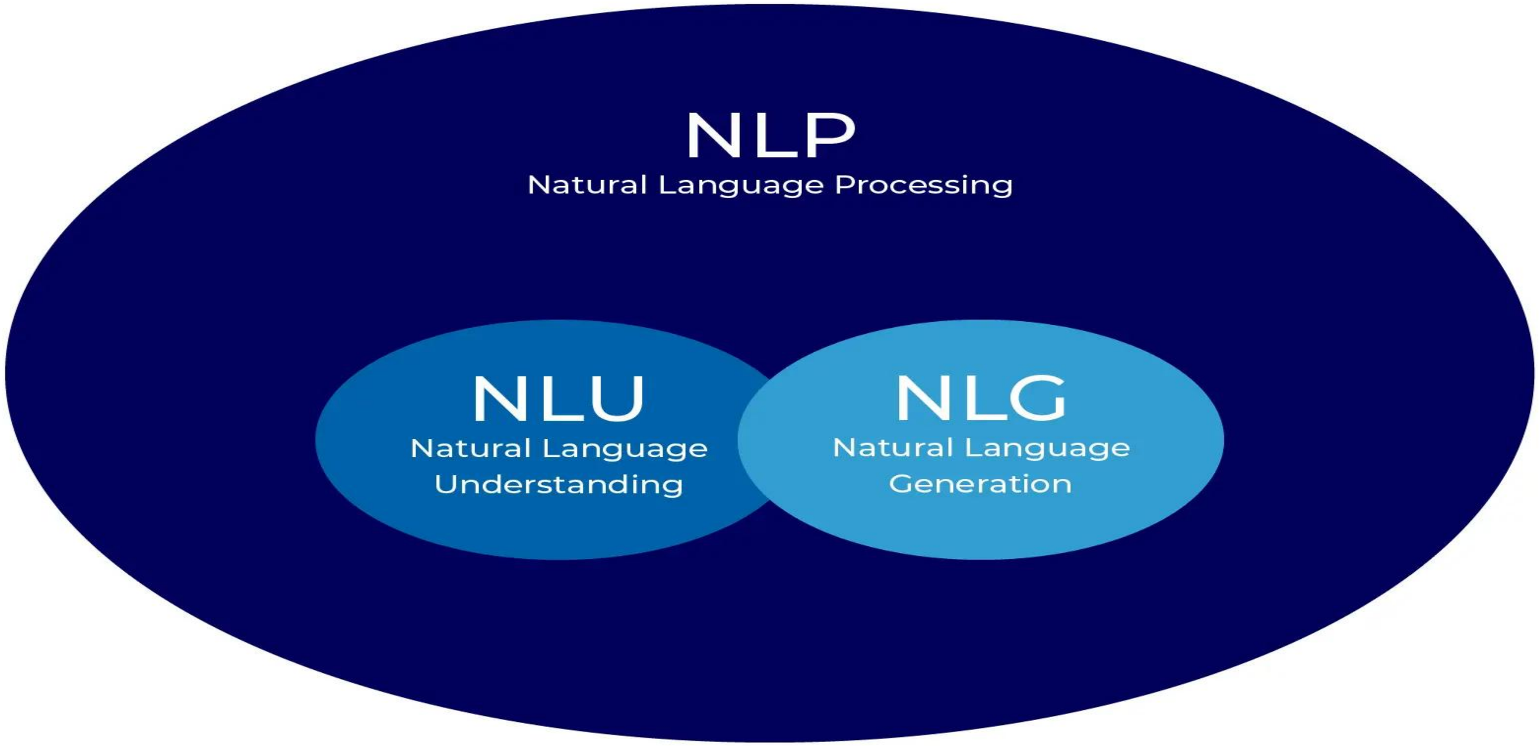
NLP Applications

NLP=NLU+NLG



NLP Applications: Speech Recognition

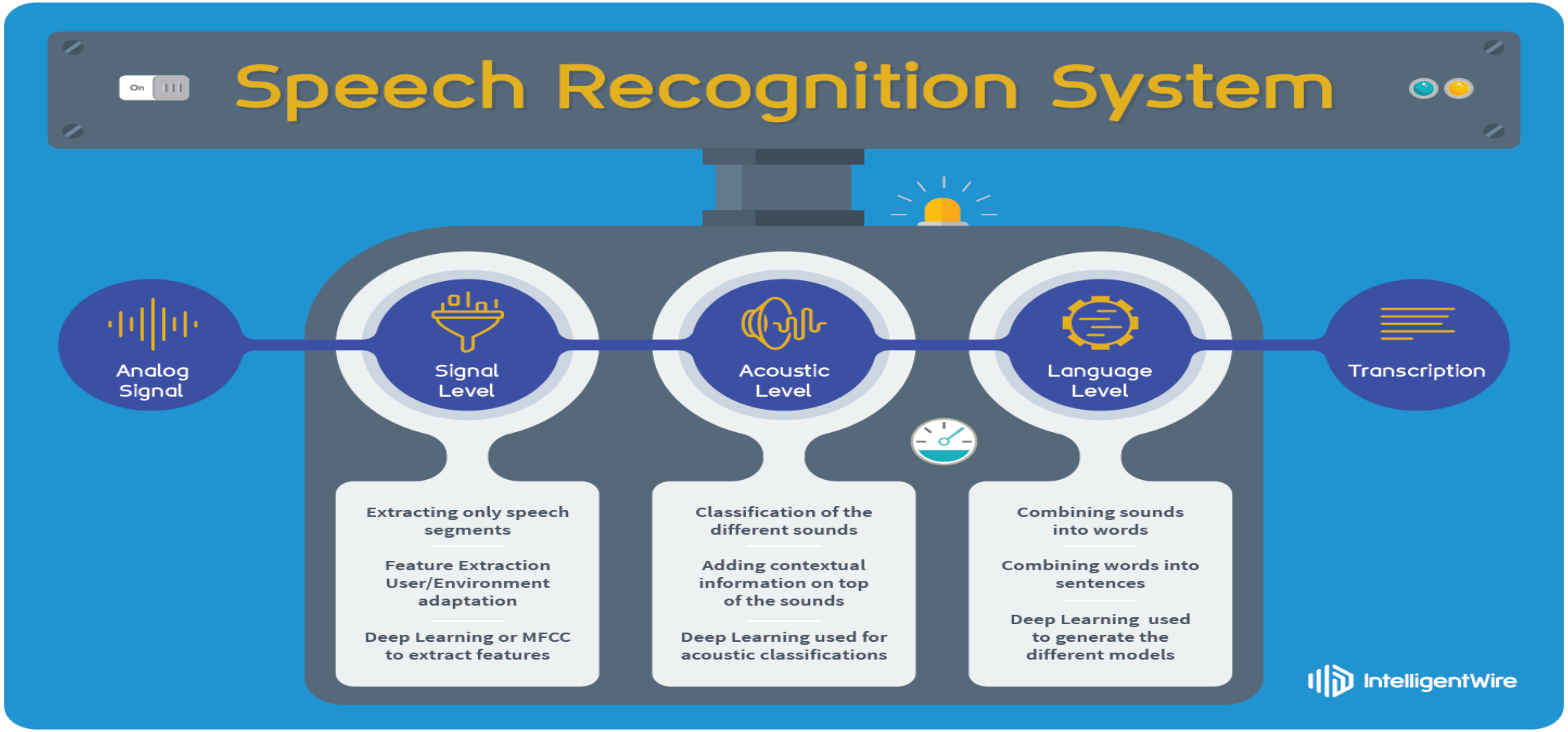
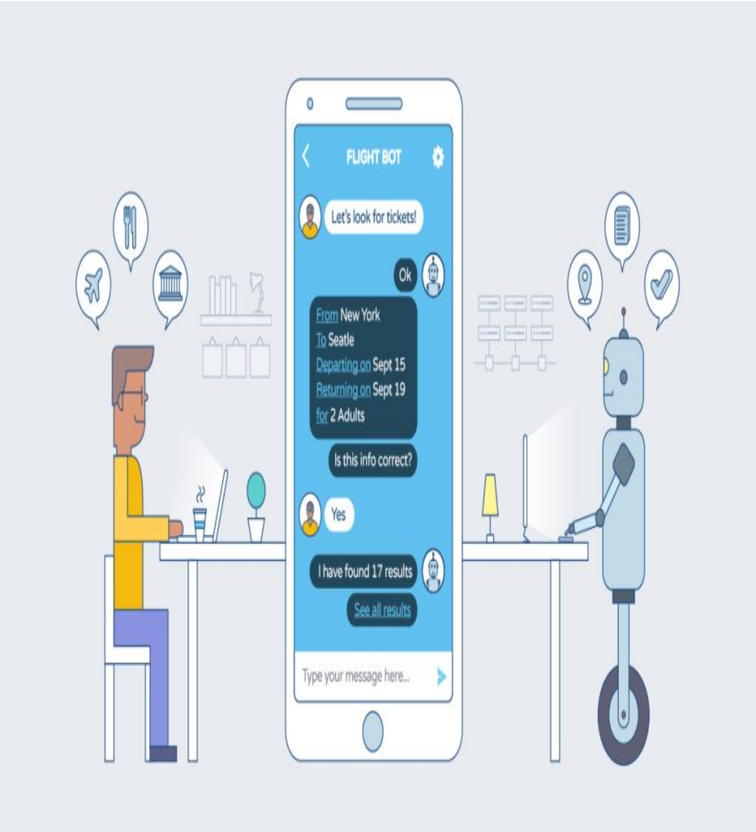
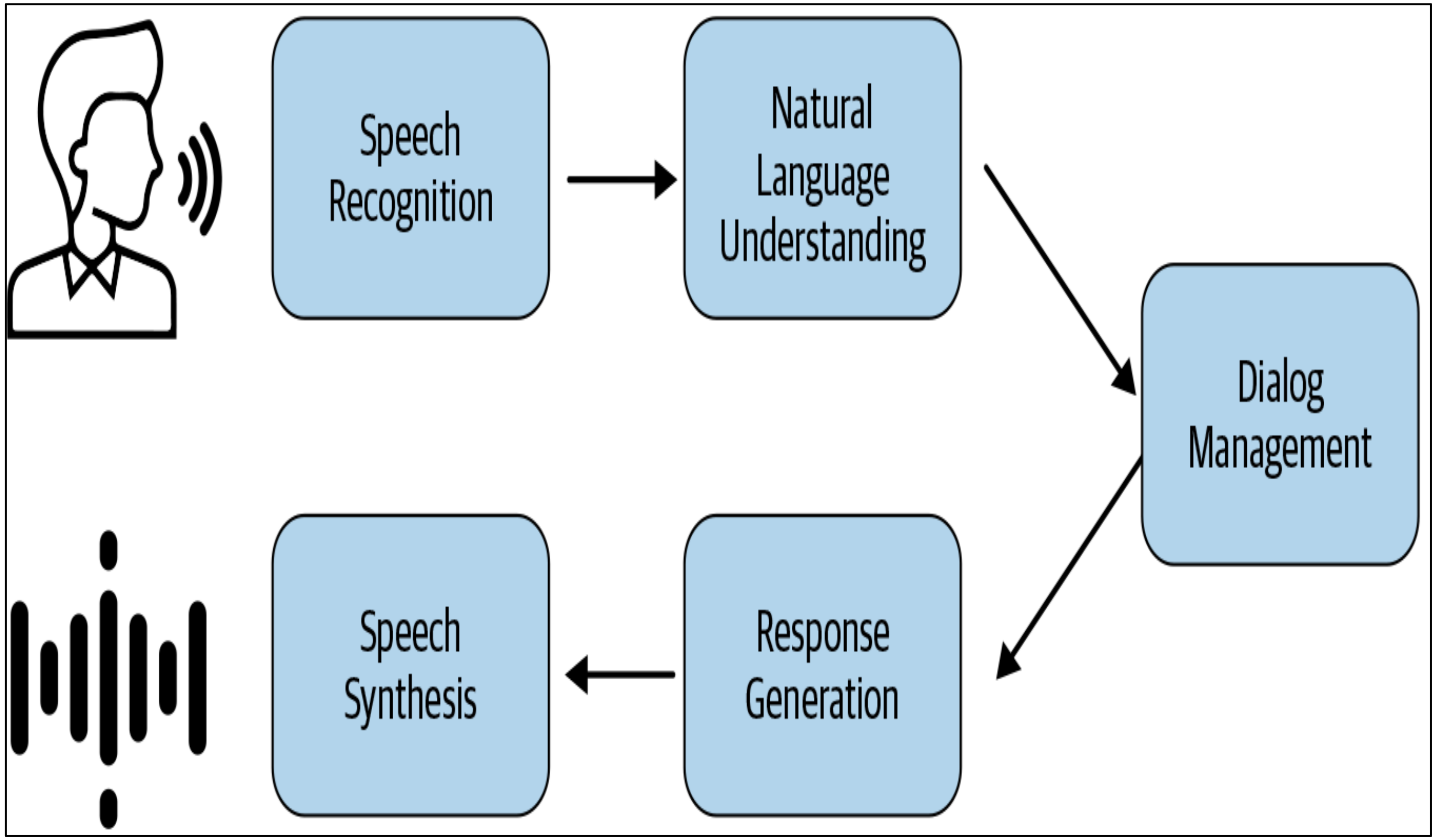


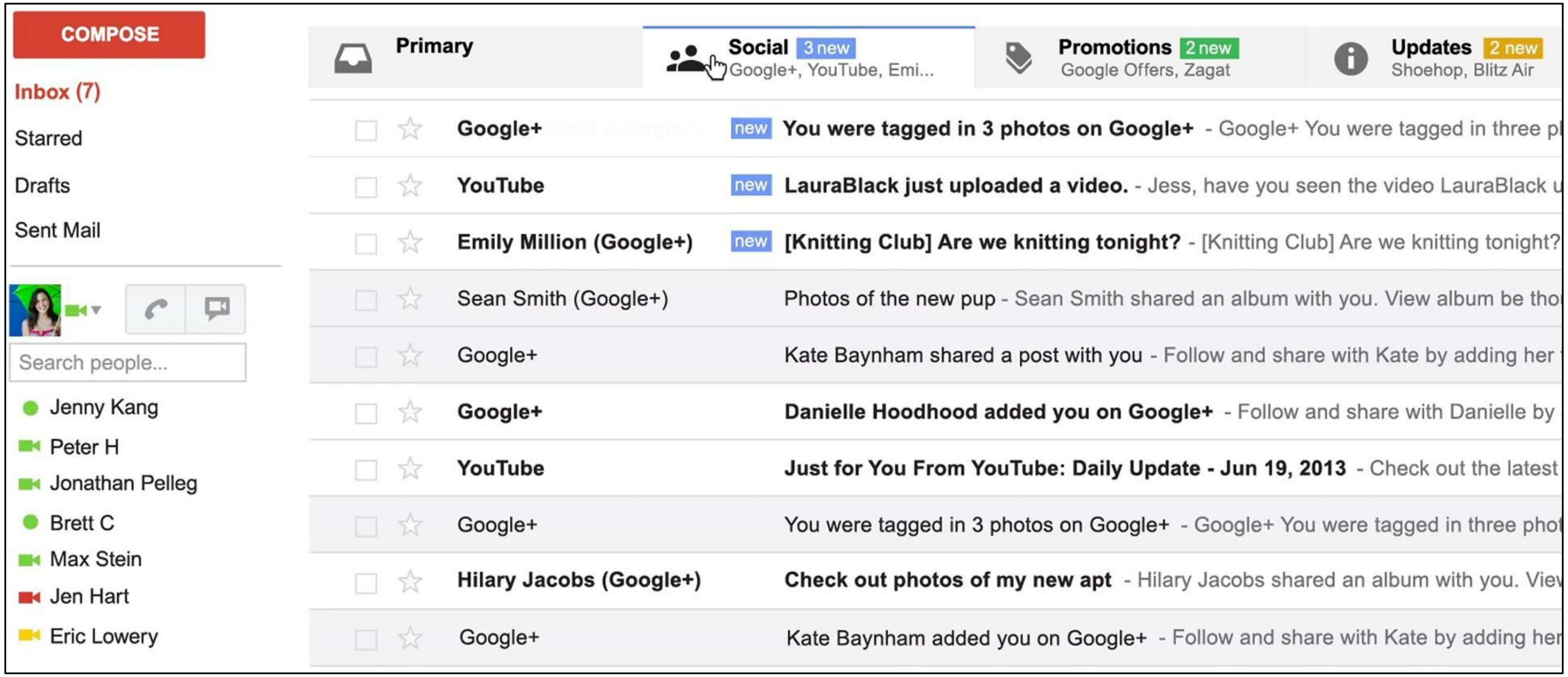
Image source: https://www.phonon.io/understanding-nlp-speech-recognition/

NLP Applications: Conversation Agents and Chatbots



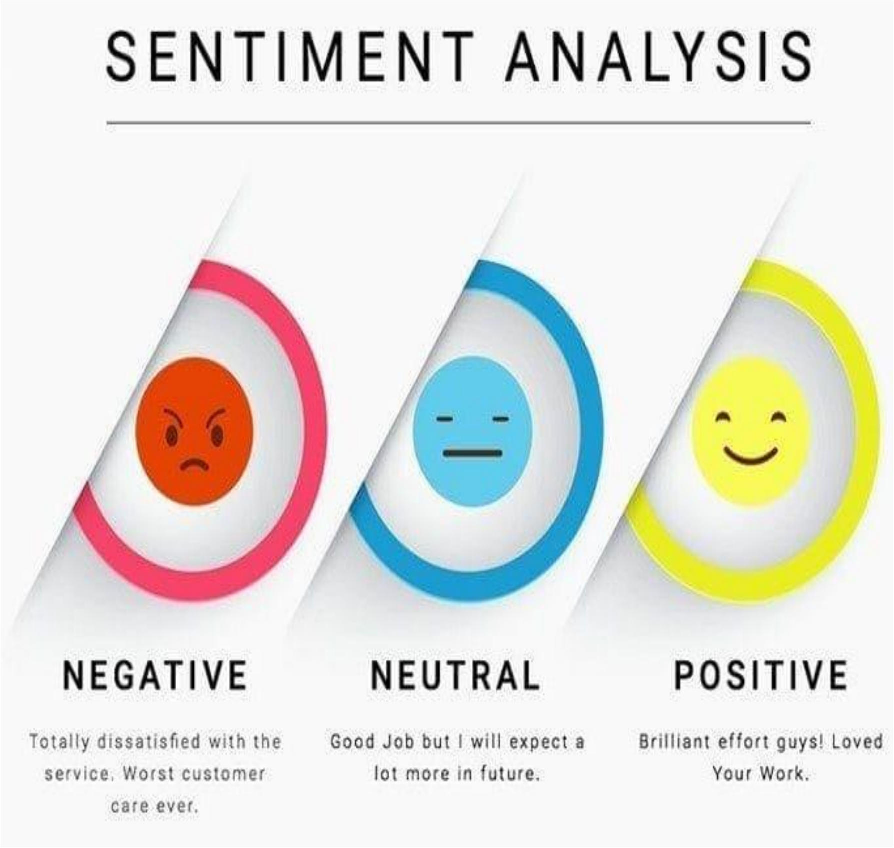


NLP Applications :Text Classification



- spam / not spam
- priority level
- category (primary / social / promotions / updates)

NLP Applications :Sentiment Analysis



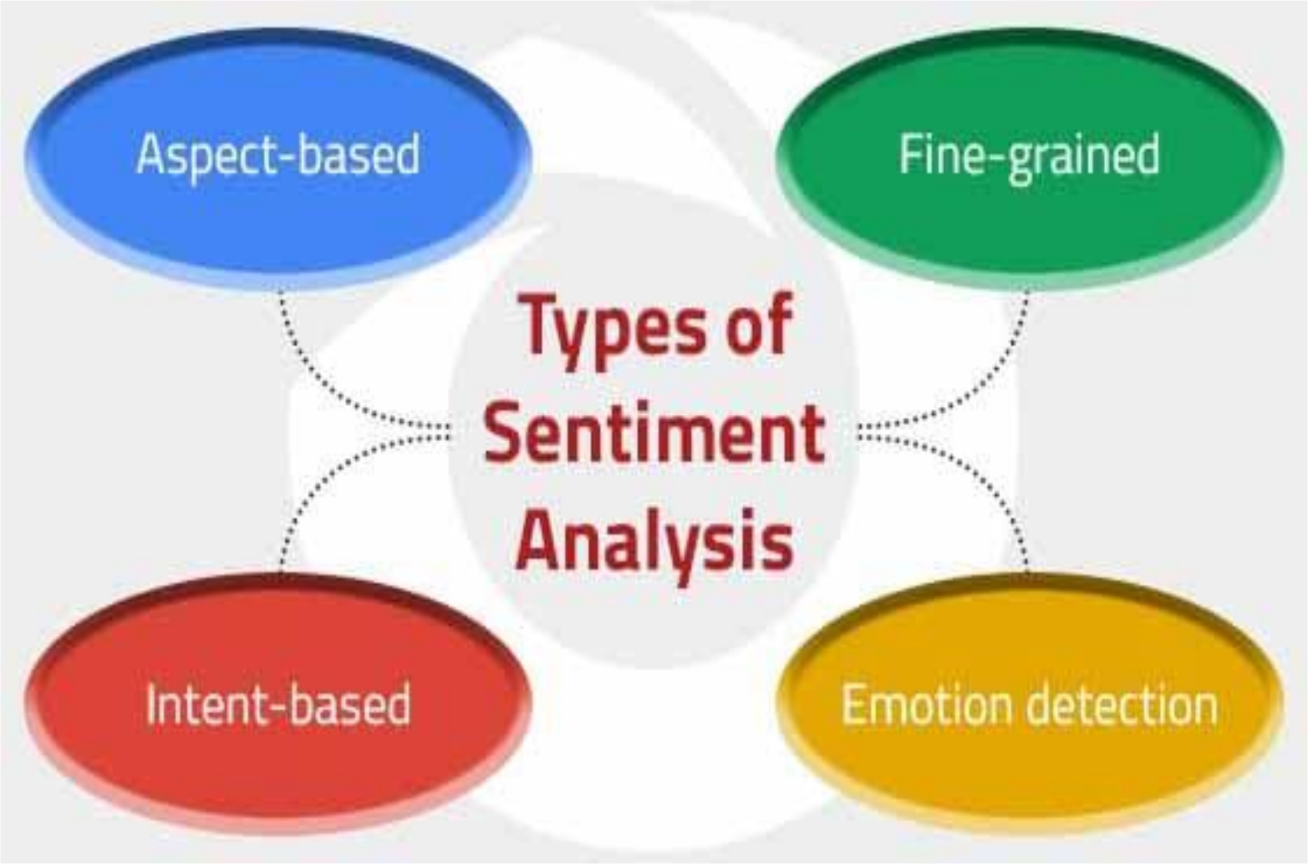
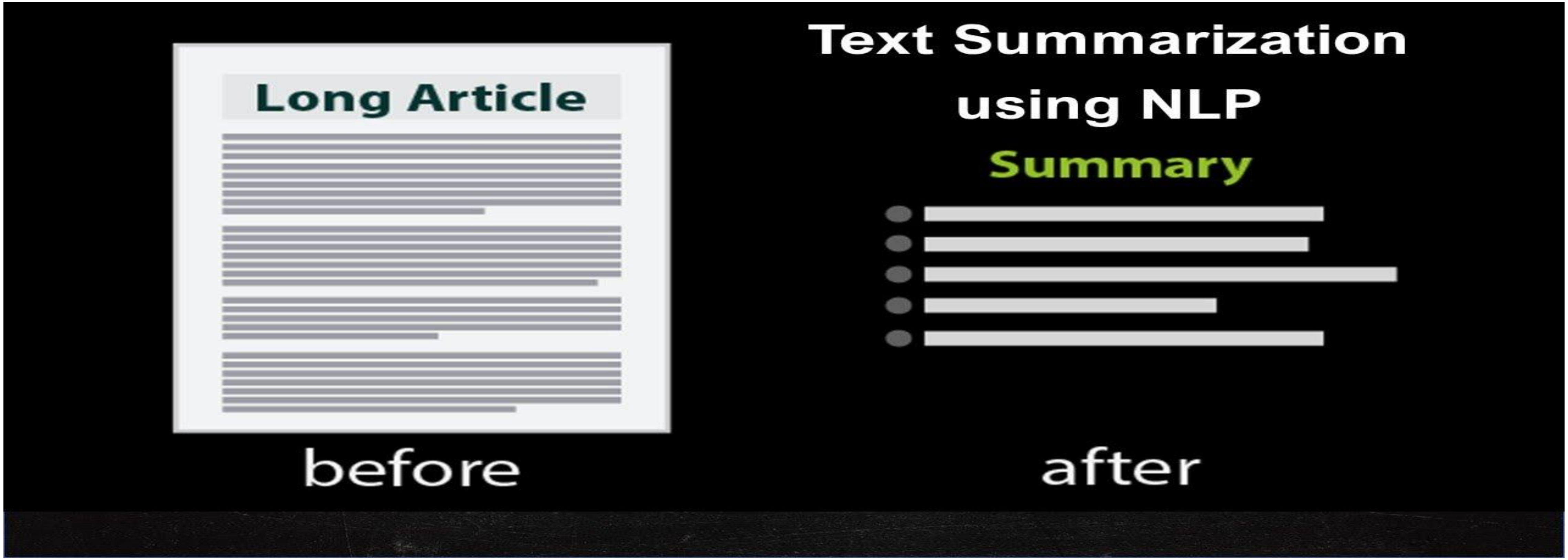
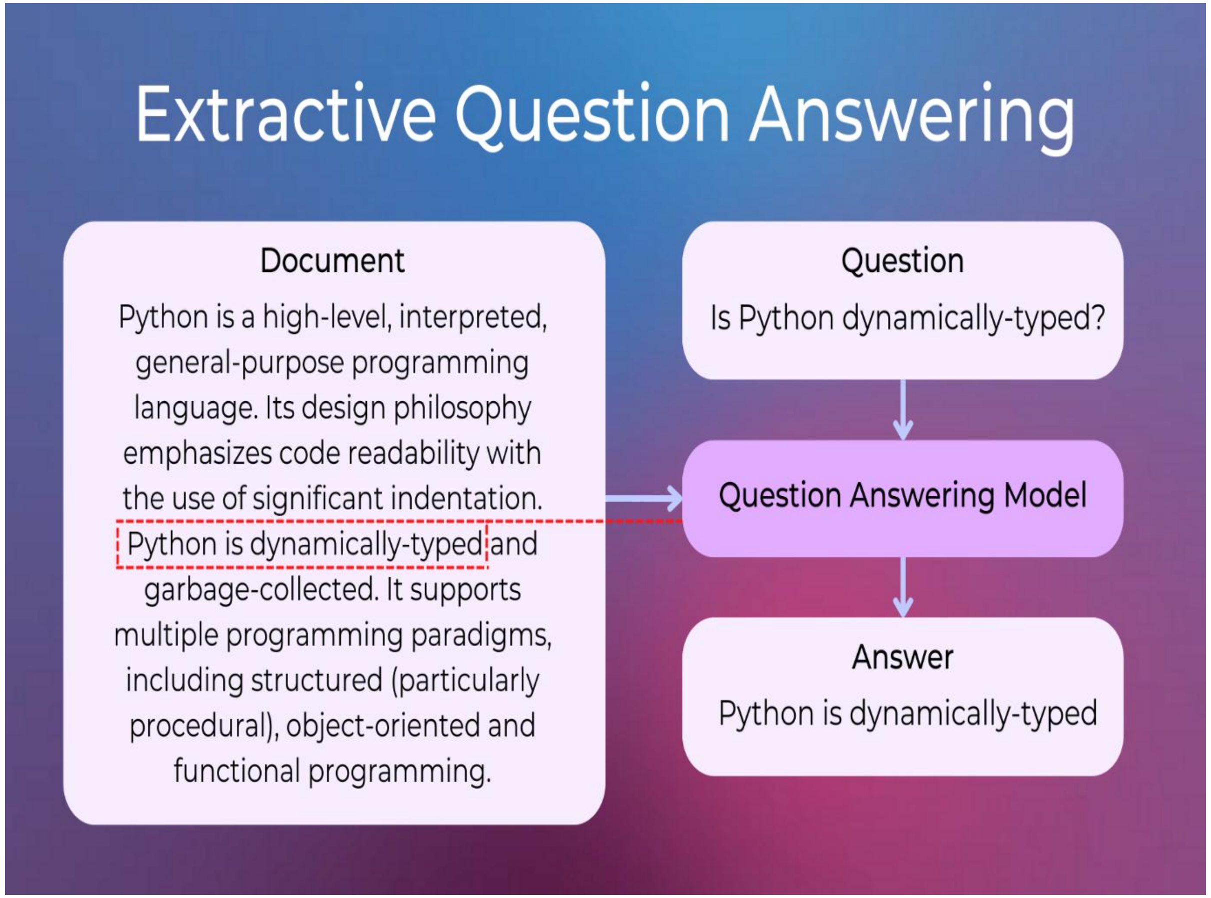


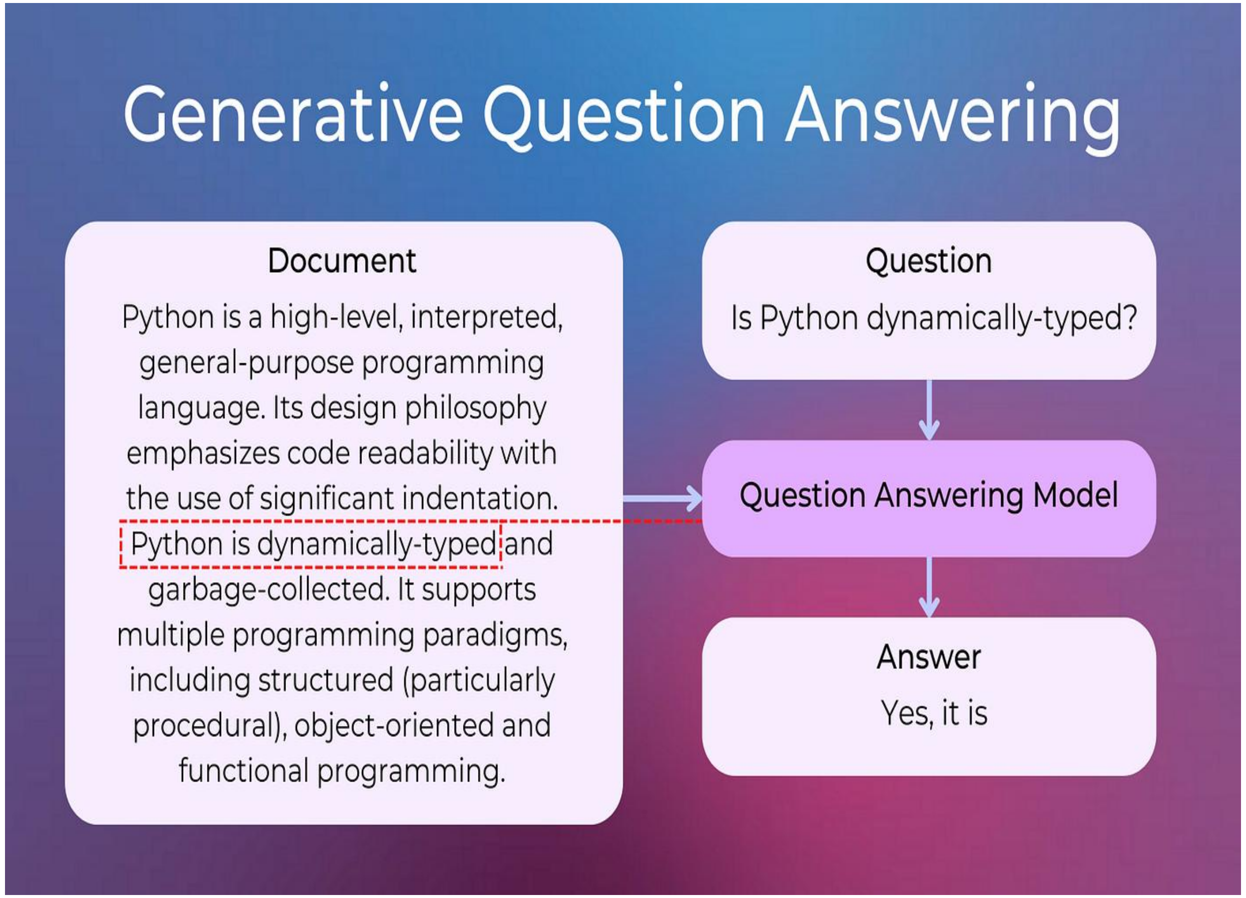
Image source: text book

NLP Applications :Text Summarization

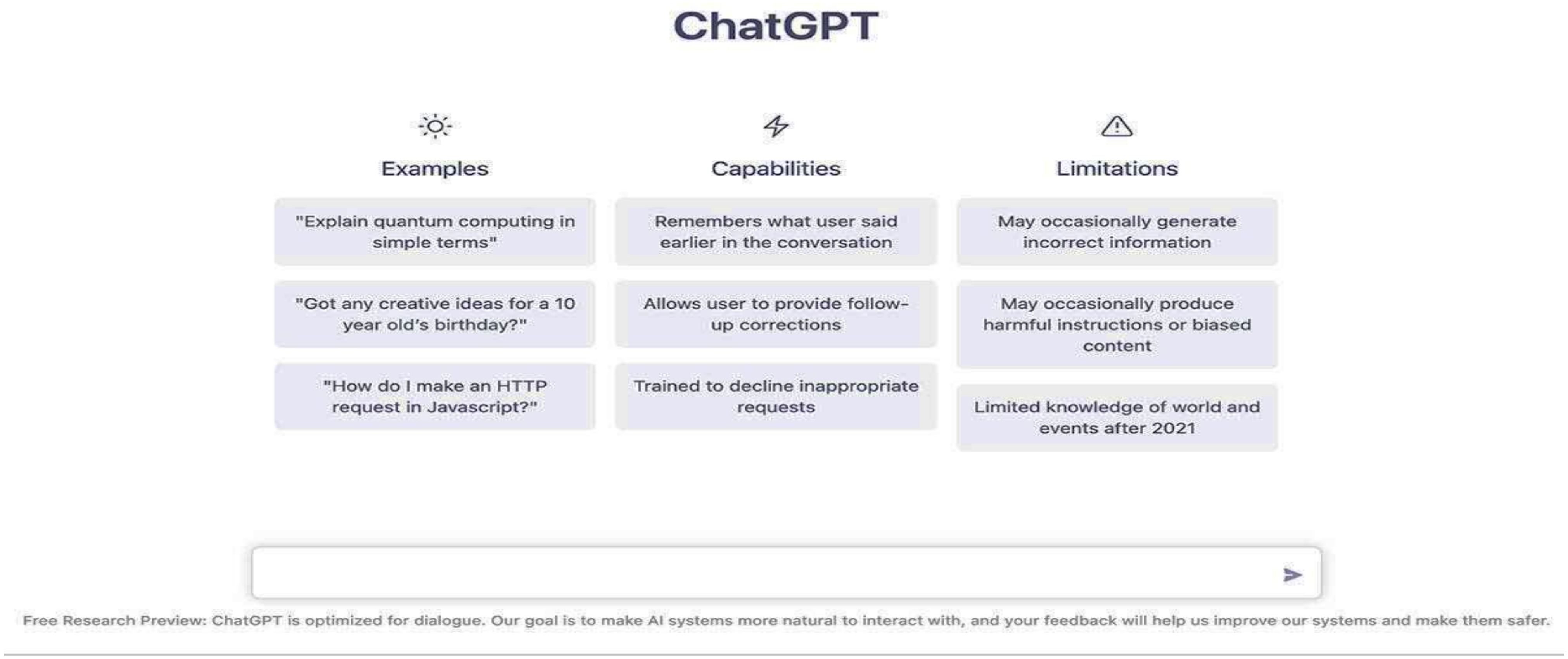


NLP Applications :Question Answering





Generative AI: OpenAI’s ChatGPT



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Challenges in NLP

- Ambiguity
- Sparsity
- Variation
- Common knowledge

Language is Ambiguous :Words Have Many Meanings

- word "bass" can refer to a type of fish or a low-frequency sound.
- The word "bank" can refer to a financial institution or the edge of a river.

Language is Ambiguous :Attachment Ambiguities

She saw the man with the telescope.

Language is Ambiguous :Coreference Ambiguities

My girlfriend and I met my lawyer for a drink

but **she** became ill and had to leave.

Challenges in NLP

- Ambiguity
- Sparsity
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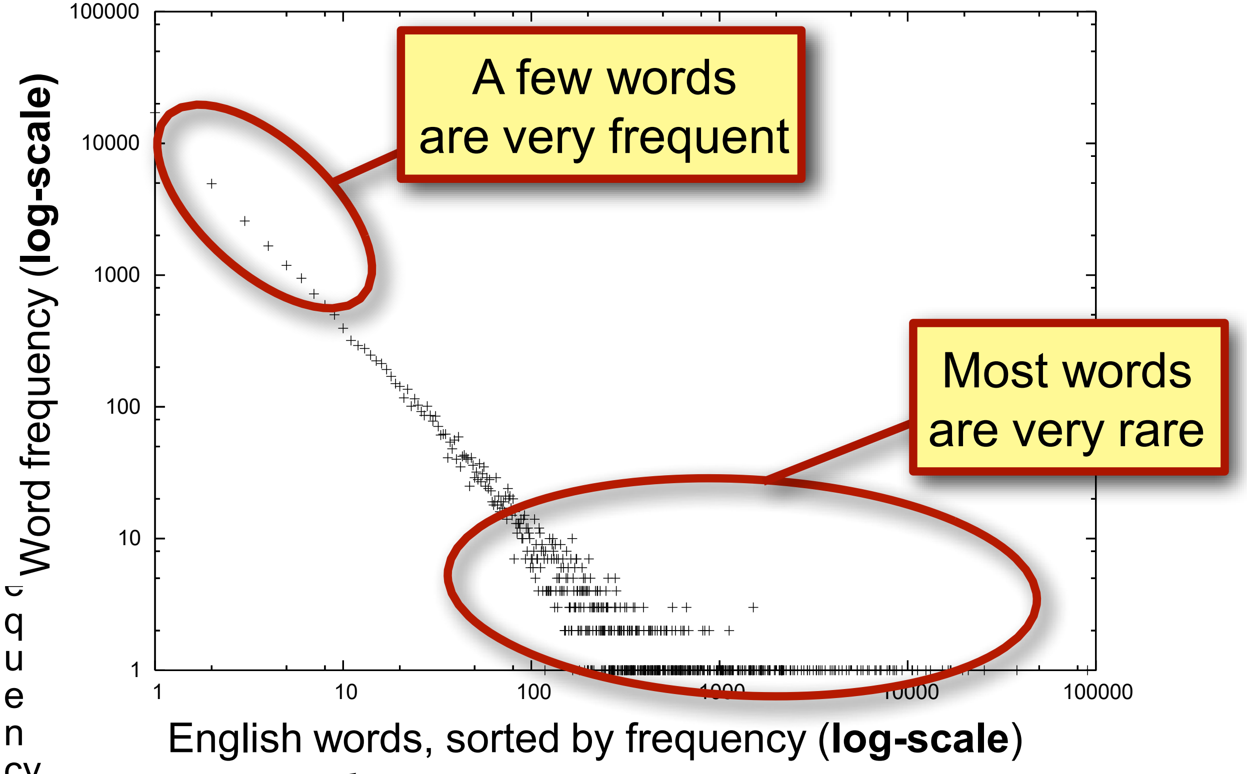
- Common knowledge

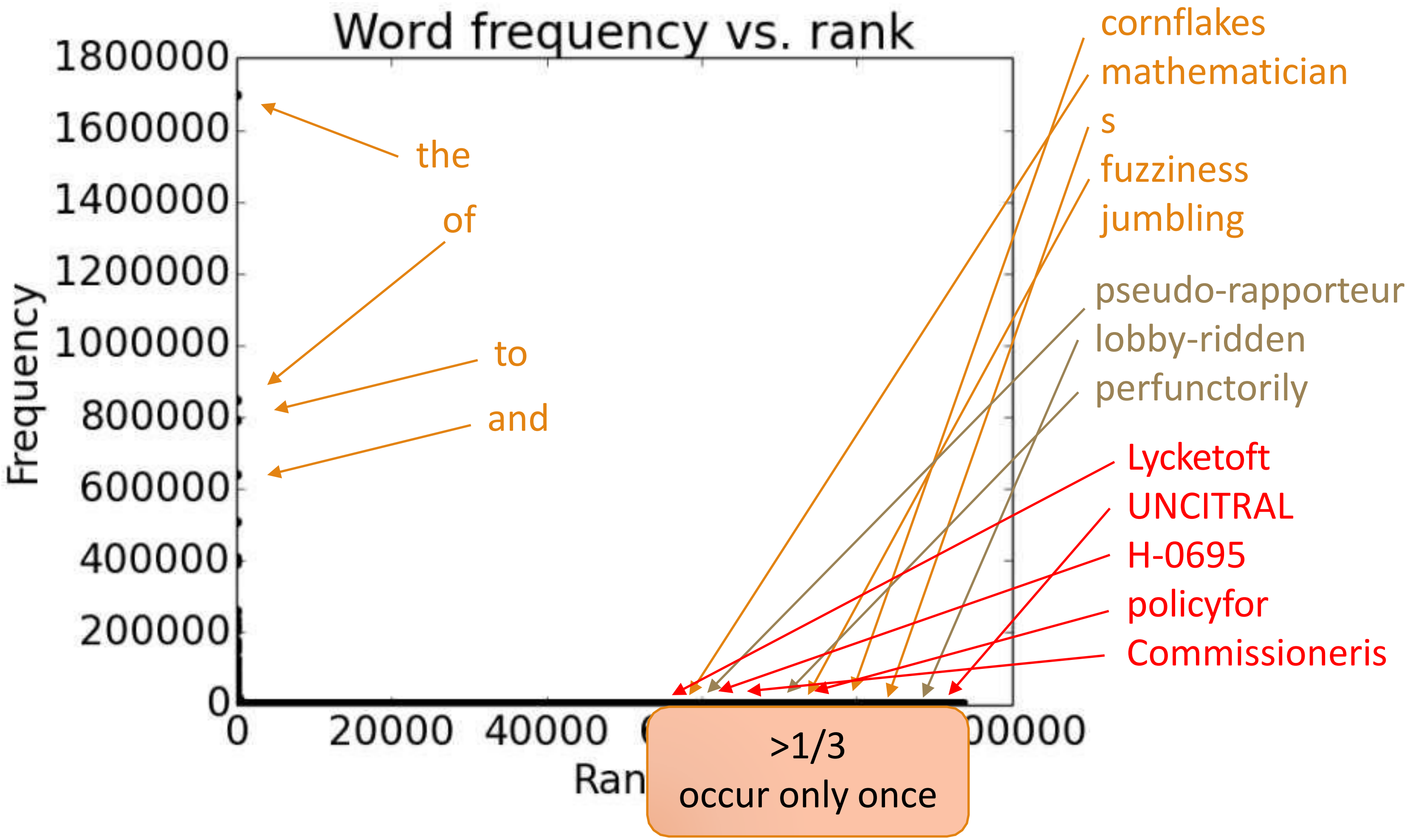
Zipf's law: the long tail

How many words occur once, twice, 100 times, 1000 times?



w_1 = *the*, w_2 = *to*, …., w_{5346} = *computer*, ...

Sparsity



Word frequency vs. rank
Frequency
1800000
1600000
1400000
1200000
1000000
800000
600000
400000
200000
0
0
20000
40000
the
of
to
and
cornflakes
mathematician
s
fuzziness
jumbling
pseudo-rapporteur
lobby-ridden
perfunctorily
Lycketoft
UNCITRAL
H-0695
policyfor
Commissioneris
>1/3
occur only once

Challenges in NLP

- Ambiguity
- Sparsity
- Variation
- Common knowledge

Variation within the same language

- ***Variation within the same language***
  - Lexical variation
    - She gave the book to Tom **vs**. She gave Tom the book
  - Regional or Geographic Variation
  - Social Variation
  - Stylistic Variation
  - Generational Variation
- ***Variation between different languages***

Challenges in NLP

- Ambiguity
- Sparsity
- Variation
- Common knowledge

Common knowledge

It is the set of all facts that most humans are aware of

- man bit dog
- dog bit man

Image generated by ChatGPT

Other Challenges ?

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#3209 585

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Approaches to NLP

- **Heuristics-Based NLP**
  - Regular Expression
- **Machine Learning for NLP**
  - Supervised
  - Unsupervised
- **Deep Learning for NLP**
  - Recurrent neural networks
  - Long short-term memory
- **Transformers**

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NLP Libraries in Python

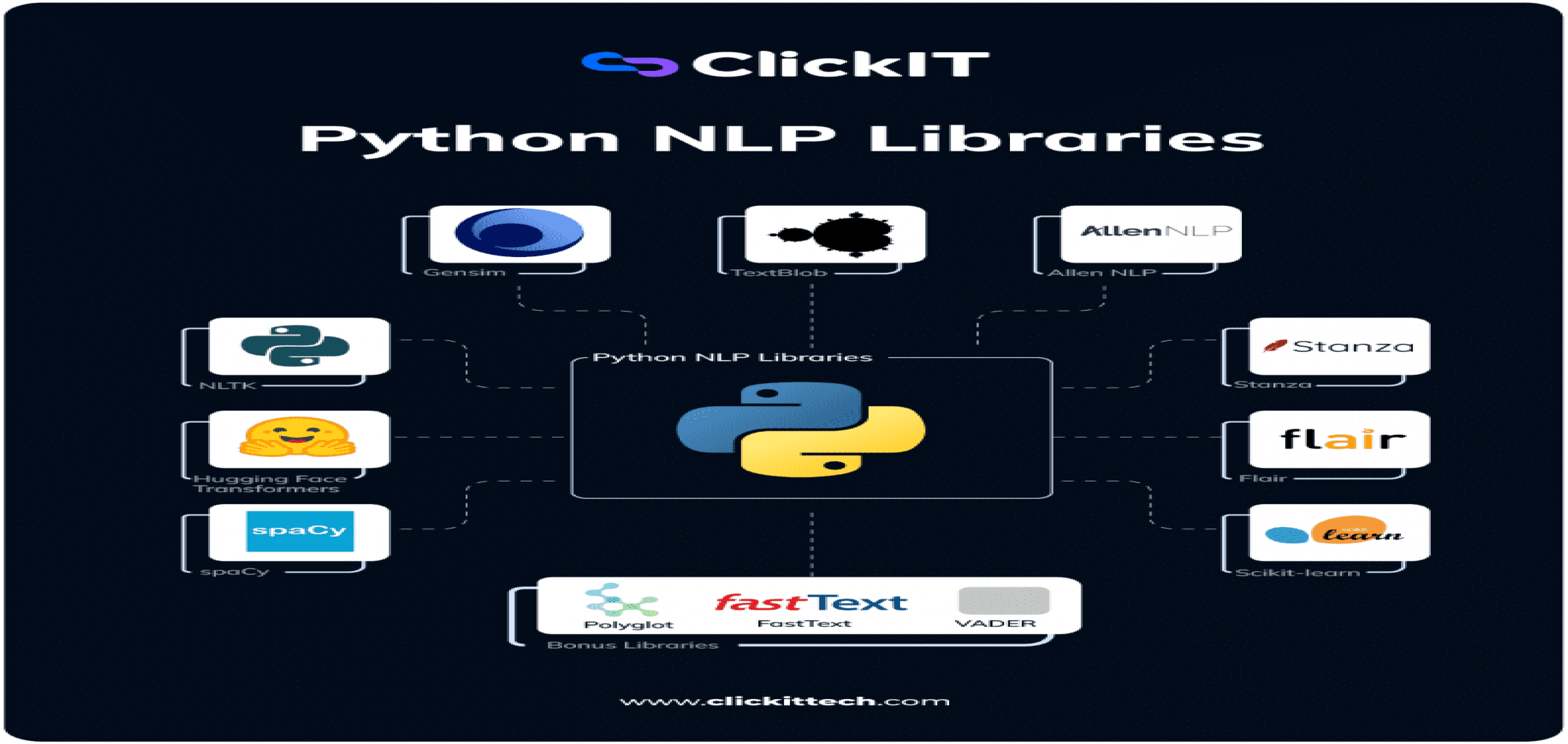


Image source: https://www.appventurez.com/blog/beginners-guide-to-natural-language-processing-nlp/

Summary

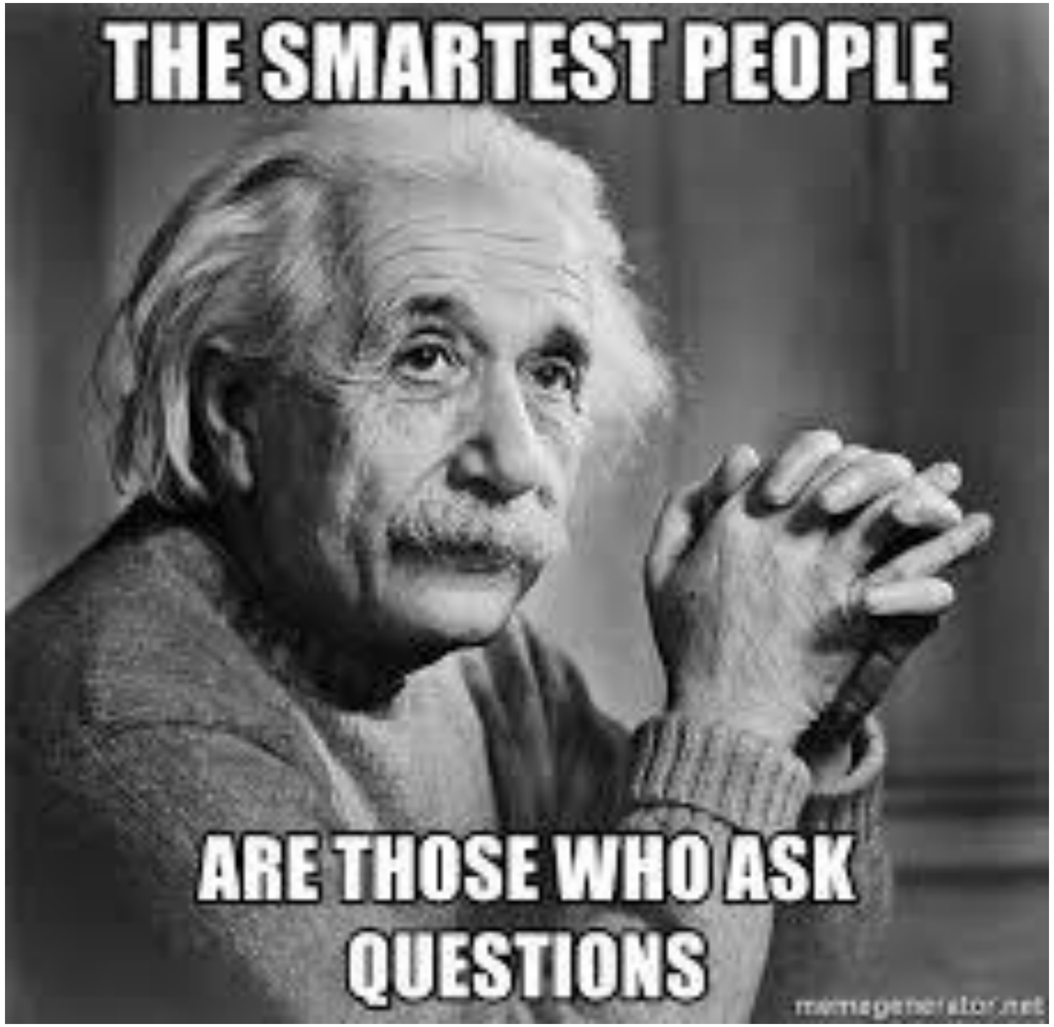
- Language to Knowledge
- Clearing the Confusion: AI vs. Machine Learning vs. Deep Learning Differences
- Lots of applications…
- It's quite difficult
- Varied, sparse, and lots of ambiguities

What's Next?

Text preprocessing and exploratory analysis

- Regular expression
- Tokenization
- Stemming
- Removing stop words
- Part Of Speech tagging (POS)

Q&A


THE SMARTEST PEOPLE
ARE THOSE WHO ASK QUESTIONS
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